

Lecture Notes

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Textbook: A Course in Linear Algebra by David B. Damiano and John B. Little

1 Vector Space

1.1 Axioms of a Field

Definition 1.1.1 (Binary Operations). A binary operation is a function $\boxplus : X \times X \rightarrow X$ with two inputs and one output.

Examples.

1. On real numbers $+$: $\mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}$.
2. On real numbers $\cdot$: $\mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}$.

Definition 1.1.2 (Associativity and Commutativity). An operation $\boxplus : X \times X \rightarrow X$ is *commutative* if $x \boxplus y = y \boxplus x$. An operation $\boxplus : X \times X \rightarrow X$ is *associative* if $x \boxplus (y \boxplus z) = (x \boxplus y) \boxplus z$.

Counterexample. Subtraction in reals: $3 - 4 \neq 4 - 3 \wedge (1 - 2) - 3 \neq 1 - (2 - 3)$.

Axiom 1.1.3 (Field). A *field* is a set $\mathbb{F}$ together with two binary operations

$$\boxplus : \mathbb{F} \times \mathbb{F} \rightarrow \mathbb{F} \quad \boxdot : \mathbb{F} \times \mathbb{F} \rightarrow \mathbb{F}$$

$A_{\mathbb{F}}1$. $\boxplus$ and $\boxdot$ are both *associative*.

$A_{\mathbb{F}}2$. $\boxplus$ and $\boxdot$ are both *commutative*.

$A_{\mathbb{F}}3$. There is an *additive identity* $\hat{0} \in \mathbb{F}$ and *multiplicative identity* $1 \in \mathbb{F}$ s.t.

$$\hat{0} \boxplus x = x \text{ and } 1 \boxdot x = x, \forall x \in \mathbb{F}.$$

$A_{\mathbb{F}}4$. For each $x \in \mathbb{F}$ there is an *additive inverse* $x' \in \mathbb{F}$ s.t.

$$x + x' = \hat{0}.$$

$A_{\mathbb{F}}5$. For each $x \neq \hat{0} \in \mathbb{F}$ there is a *multiplicative inverse* $x' \in \mathbb{F}$ such that

$$x \boxdot x' = 1$$

$A_{\mathbb{F}}6$. The multiplication operator is *distributive* over the addition operator: $a \boxdot (b \boxplus c) = a \boxdot b \boxplus a \boxdot c$.

Remarks 1.1.3.0.1. Two common fields are $\mathbb{C}$ and $\mathbb{R}$. Also, two common nonexamples include $\mathbb{N}$ and $\mathbb{Z}$. Since for naturals, $A_{\mathbb{F}}4$. is not satisfied; for integers, $A_{\mathbb{F}}5$. is not satisfied. Furthermore, the order of the axioms matters.

1.2 Complex Numbers

Definition 1.2.1 (The Imaginary Number i). The imaginary number i is defined:

$$i^2 = -1 \iff i = \sqrt{-1}.$$

Definition 1.2.2 (Complex Numbers). Complex numbers are numbers of the form: $z = a + bi$ where $a, b \in \mathbb{R}$, and $a = \operatorname{Re}(z)$ is the real part of the complex number z and $b = \operatorname{Im}(z)$ is the imaginary part of z .

Theorem 1.2.2.1 (Subtraction and Division). Let $z_1, z_2 \in \mathbb{C}$ s.t. $z_1 = a + bi$ and $z_2 = c + di$ for some $a, b, c, d \in \mathbb{R}$ and imaginary number i . Then,

1.

$$z_1 - z_2 = (a - c) + (b - d)i.$$

2.

$$\frac{z_1}{z_2} = \frac{ac + bd}{c^2 + d^2} + \frac{bc - ad}{c^2 + d^2}i.$$

Proof.

$$\begin{aligned} z_1 - z_2 &= z_1 + (-1)z_2 \\ &= (a + bi) + (-1)(c + di) \\ &= a + bi - c - di \\ &= (a - c) + (b - d)i \end{aligned} \quad \text{as required.}$$

□

Proof.

$$\begin{aligned} \frac{z_1}{z_2} &= \frac{a + bi}{c + di} \\ &= \frac{a + bi}{c + di} \cdot \frac{c - di}{c - di} \\ &= \frac{ac - adi + bci + bd}{c^2 + d^2} \\ &= \frac{ac + bd}{c^2 + d^2} + \frac{bc - ad}{c^2 + d^2}i \end{aligned} \quad \text{as required.}$$

□

Definition 1.2.3 (Complex Conjugate). Let $z = a + bi$. Then, the complex conjugate of z , denoted z^* or $\bar{z}$ is defined:

$$z^* = a - bi.$$

Definition 1.2.4 (Modulus $|z|$). The modulus of a complex number z (analogously absolute value in reals) is defined:

$$|z|^2 = zz^* = a^2 + b^2 \iff |z| = \sqrt{zz^*} = \sqrt{a^2 + b^2}.$$

Theorem 1.2.4.1 (Division). Let $z_1, z_2 \in \mathbb{C}$ s.t. $z_1 = a + bi$ and $z_2 = c + di$ for some $a, b, c, d \in \mathbb{R}$ and imaginary number i . Then,

$$\frac{z_1}{z_2} = \frac{z_1 z_2^*}{|z_2|^2}.$$

Proof.

$$\begin{aligned} \frac{z_1}{z_2} &= \frac{ac + bd}{c^2 + d^2} + \frac{bc - ad}{c^2 + d^2}i && \text{by thm 1.2.2.1.} \\ &= \frac{(a + bi)(c - di)}{c^2 + d^2} && \text{algebra} \\ &= \frac{z_1 z_2^*}{|z_2|^2} && \text{as required.} \end{aligned}$$

□

1.3 Vectors and Vector Space

Axiom 1.3.1 (Vector Space). A vector space is a set V with two binary operations:

$$\boxplus : V \times V \rightarrow V \quad \boxdot : \mathbb{F} \times V \rightarrow V$$

called vector addition and vector multiplication such that the following axioms hold:

$A_{\forall}1.$ $\boxplus$ is *associative*.

$A_{\forall}2.$ $\boxplus$ is *commutative*.

$A_{\forall}3.$ There is an *additive element* $\vec{0} \in V$ s.t.

$$\vec{0} \boxplus \vec{x} = \vec{x}$$

for all $\vec{x} \in V$. We call $\vec{0}$ the *zero vector* of V .

$A_{\forall}4.$ For each $\vec{x} \in V$, there is an additive inverse $\vec{x}' \in V$ such that

$$\vec{x} \boxplus \vec{x}' = \vec{0}.$$

$A_{\forall}5.$ For all $\vec{x}$ and $\vec{y}$ and $c \in \mathbb{F}$, scaling by c *distributes over* addition:

$$c \boxdot (\vec{x} \boxplus \vec{y}) = c\vec{x} \boxplus c\vec{y}.$$

$A_{\forall}6.$ For all $\vec{x} \in V$ and $c, d \in \mathbb{F}$, the *field addition* in $\mathbb{F}$ *distributes scalar multiplication* in V :

$$(c + d) \boxdot \vec{x} = (c \boxdot \vec{x}) \boxplus (d \boxdot \vec{x}).$$

$A_{\forall}7.$ For all $\vec{x} \in V$ and $c, d \in \mathbb{F}$, $\boxdot$ associates with scalar multiplication on $\mathbb{F}$:

$$(cd) \boxdot \vec{x} = c \boxdot (d \boxdot \vec{x}).$$

$A_{\forall}8.$ For all $\vec{x} \in V$, we have $1 \boxdot \vec{x} = \vec{x}$, the 1 is the *multiplicative identity* from $\mathbb{F}$.

Proposition 1.3.1.1. Let V be a vector space. Then,

$$\left\{ \begin{array}{l} i., 0 \text{ is unique.} \\ ii., \forall x \in V, 0x = 0. \\ iii., \text{for each } x \in V, -x \text{ is unique} \\ iv., \forall x \in V, \forall c \in \mathbb{R}, (-c)x = -(cx) \end{array} \right.$$

The proof is given on page 10 of the textbook.

1.4 Subspaces

Definition 1.4.1 (Subspace). Let V be a vector space with $W \subseteq V$. Then W is a (vector) subspace iff W is a vector space itself under the operation of vector sum and scalar multiplication from V .

Theorem 1.4.1.1 (Closure of Linear Combination). Let V be a vector space with $W \neq \emptyset \wedge W \subseteq V$. Then W is a subspace of V iff $\forall x, y \in W, \forall c \in \mathbb{R}, cx + y \in W$.

Remarks 1.4.1.1.1. By Definition 1.1.1, the additive identity must be contained in a vector space, hence it is necessary for W to be a non-empty set.

Proof.

$\implies$: Suppose W is a subspace of V . Then, W is both *closed under addition* and *closed under scalar multiplication*. That is, first, $\forall x \in W, \forall c \in \mathbb{R} cx \in W$. Furthermore, $\forall y \in W, y + cx \in W$ (since cx is an element of W as W is closed under scalar multiplication; and $cx + y$ is but vector addition).

$\impliedby$: Suppose $\forall x, y \in W, \forall c \in \mathbb{R}, cx + y \in W$. We show that for a set W satisfying the preamble of the theorem, it is closed under scalar multiplication and addition. We consider three cases. First, suppose $c = 1$, then $x + y \in W$, i.e., W is closed under addition. Suppose $c = -1, y = x$, then $-1x + x = 0x = 0 \in W$. Suppose $y = 0$, note that we have established that $0 \in W$, it follows that $cx \in W$, i.e., W is closed under multiplication. W is thus by definition a vector space and furthermore under the operation of vector sum, i.e., W is a subspace of V as required. $\square$

Application. Consider $W \subseteq V \in \mathbb{R}^3$ s.t. $W = \{(x_1, x_2, x_3) \mid \text{constraint .}\}$ Then we can show that W is a subspace of $\mathbb{R}^3$ by assuming $x, y \in W \wedge c \in \mathbb{R}$ such that $cx + y \in W$ is satisfied.

Theorem 1.4.1.2 ($W_1 \cap W_2$ is a subspace). Let V be a vector space. Then, the intersection of any collection of subspaces of V is a subspace of V .

Notations. a_{ij} refers to the coefficient of x_j in i^{th} equation. For example, a_{23} is the coefficient of x_3 in the second equation of a *simultaneous system*.

Corollary 1.4.1.2.1. Let $a_{ij} (1 \leq i \leq m, 1 \leq j \leq n)$ be any real number and let $W = \{(x_1, \dots, x_n) \in \mathbb{R}^n \mid a_{i1}x_1 + \dots + a_{in}x_n = 0 \forall i \text{ s.t. } 1 \leq i \leq m.\}$ Then, W is a subspace of $\mathbb{R}^n$.

1.5 Linear Combinations

Definition 1.5.1 (Linear Combination). For a set $S \subseteq V$, we define a *linear combination* of vectors from S to be:

$$a_1 \mathbf{v}_1 + \cdots + a_k \mathbf{v}_k = \sum_{i=1}^k a_i \mathbf{v}_i,$$

where $\mathbf{v}_i \in V$ and $a_i \in \mathbb{F}$. We say $a_1 \mathbf{v}_1 + \cdots + a_k \mathbf{v}_k$ is a linear combination of $\{\mathbf{v}_1, \dots, \mathbf{v}_k\}$.

Remark. Note that each of vector $\mathbf{v}$ itself has the form of coordinate. Intuitively, a linear combination is combinations of elements “built” by addition and scaling of a set of vectors.

Definition 1.5.2 (Span). For a subset $S \subseteq V$ we define the *span* of S to be:

$$\text{Span}(S) = \{\text{all linear combinations of elements in } S\}.$$

Furthermore, $\text{Span}(\emptyset) = \{\mathbf{0}\}$ since the empty sum is 0.

Theorem 1.5.2.1 (Spans Are Subspaces). If $S \subseteq V$ is a subset of V , then $\text{Span}(S) \subseteq V$ is a subspace of V . Note that S need not be subspace.

Proof. We prove by cases. First, note that $S = \emptyset \vee S \neq \emptyset$. Suppose, for case 1, $S \neq \emptyset$. Then, for elements in $\text{Span}(S)$, say $\vec{x}, \vec{y} \in \text{Span}(S)$ and $c \in \mathbb{F}$, we have

$$\begin{cases} (1) & \vec{x} = a_1 \mathbf{x}_1 + \cdots + a_k \mathbf{x}_k \text{ for some } \mathbf{x}_i \in S \text{ and } a_i \in \mathbb{F} \\ (2) & \vec{y} = b_1 \mathbf{y}_1 + \cdots + b_k \mathbf{y}_k \text{ for some } \mathbf{y}_i \in S \text{ and } b_i \in \mathbb{F} \end{cases}$$

Furthermore, it follows that

$$\begin{aligned} c\vec{x} + \vec{y} &= (c(a_1 \mathbf{x}_1 + \cdots + a_k \mathbf{x}_k)) + (b_1 \mathbf{y}_1 + \cdots + b_k \mathbf{y}_k) \\ &= ca_1 \mathbf{x}_1 + \cdots + ca_k \mathbf{x}_k + b_1 \mathbf{y}_1 + \cdots + b_k \mathbf{y}_k \quad \text{by distributivity and associativity} \\ &\in \text{Span}(S) \quad \text{by definition of Span since the expression is itself a linear combination of } S \end{aligned}$$

Also, note that by definition of Span, and the fact that for any set S , $\emptyset \in S$, it is always the case that $0 \in \text{Span } S$. Thus, by closure theorem, we conclude in our first case that $\text{Span}(S)$ is indeed a subspace. Suppose, for case 2 that $S = \emptyset$. Then, $\text{Span}(S) = \text{Span}(\emptyset) = \{\vec{0}\}$ by definition. Furthermore, $\{\vec{0}\}$ is a trivial subspace. Thus, we have proven the required. $\square$

Theorem 1.5.2.2 (sum of subspaces). If $W_1, W_2 \subseteq V$ are subspaces, then $W_1 + W_2$ is also a subspace.

Theorem 1.5.2.3 (span and subspaces). If $W_1 = \text{Span}(S_1), W_2 = \text{Span}(S_2)$, then $W_1 + W_2 = \text{Span}(S_1 \cup S_2)$.

Definition 1.5.3 (Redundant Vector). The vector $\vec{w} \in \{\vec{w}, \vec{v}_1, \dots, \vec{v}_k\}$ is *redundant* if:

$$\text{Span}(\{\vec{w}, \vec{v}_1, \dots, \vec{v}_k\}) = \text{Span}(\{\vec{v}_1, \dots, \vec{v}_k\}).$$

Definition 1.5.4 (Linear Independence). We negate the definition of linear dependence and thus obtain: A subset S of a vector space V is *linearly independent* whenever we have $a_1 \in \mathbb{R} \wedge x_1 \in S$ s.t. $a + 1a_1x_1 + \dots + a_nx_n = 0$ then $a_i = 0, \forall i$:

$$\forall x \in S, \forall a_i \in \mathbb{R}, ((a_1x_1 + \dots + a_nx_n = 0) \implies \forall i \in \mathbb{R}, (a_i = 0)))$$

Theorem 1.5.4.1 (Redundant Vectors vs L.I.). If S contains a redundant vector, then S is linearly dependent. The contrapositive follows: if S is linearly independent, then S contains no redundant vectors.

Theorem 1.5.4.2 (Independence and Uniqueness of LC). Let $S = \{\vec{v}_1, \dots, \vec{v}_n\}$ be a set of vectors. The following are equivalent:

- i. S is linearly independent
- ii. $\mathbf{0} \in \text{Span}(S)$ is represented uniquely
- iii. Every vector in $\text{Span}(S)$ is represented uniquely.

Definition 1.5.5 (Linear Independence). We negate the definition of linear dependence and thus obtain: A subset S of a vector space V is *linearly independent* whenever we have $a_1 \in \mathbb{R} \wedge x_1 \in S$ s.t. $a + 1a_1x_1 + \cdots + a_nx_n = 0$ then $a_i = 0, \forall i$:

$$\forall x \in S, \forall a_i \in \mathbb{R}, ((a_1x_1 + \cdots + a_nx_n = 0) \implies \forall i \in \mathbb{R}, (a_i = 0))$$

Proposition 1.5.5.1. Let S, S' be subset of a vector space V . Let S be linearly dependent on V with $S \subseteq S'$, then S is also linearly dependent.

Proof. Intuitively, it is given that $\exists x_i \in S \exists a_i \in \mathbb{R}$ s.t. $((a_1x_1 + \cdots + a_nx_n = 0) \wedge \exists i(a_i \neq 0))$. Since $S \subseteq S'$, it must be the case that such existence in S is contained in its superset S' . $\square$

Proposition 1.5.5.2. Let S, S' be subset of a vector space V . Let S be linearly independent on V with $S \supseteq S'$, then S' is also linearly independent.

Proof. By definition $\forall x_i \in S \forall a_i \in \mathbb{R} ((a_1x_1 + \cdots + a_nx_n = 0) \implies \forall i, (a_i = 0))$. Since $S' \subseteq S$, for all elements in S' the satisfactory condition for linear independent hold. $\square$

1.6 Interlude on Solving System of Linear Equation

Definition 1.6.1 (System of Linear Equation). A system of m equations in n unknowns $x_1, \dots, x_n$ of the form:

$$\begin{aligned} a_{11}x_1 + \dots + a_{1n}x_n &= b_1 \\ a_{21}x_1 + \dots + a_{2n}x_n &= b_2 \\ &\vdots \\ a_{m1}x_1 + \dots + a_{mn}x_n &= b_m \end{aligned}$$

where $a_{ij} \in \mathbb{R} \wedge b_i \in \mathbb{R}$ is called a *system of linear equation*. a_i are called the *coefficient* of the system. The system is said to be *homogeneous* if all $b_i = 0$, and *inhomogeneous* otherwise. A *solution* (vector) of the system is a vector $(x_1, \dots, x_n) \in \mathbb{R}^n$ whose components solve all the equations in the system. A homogeneous system always has a *trivial solution* $(0, \dots, 0)$.

Definition 1.6.2 (Equivalence). Two system of linear equations are said to be *equivalent* if their sets of solutions are the same (i.e., all solutions in A are exactly in B and vice-versa).

Proposition 1.6.2.1 (Elementary Operation). Consider a system of linear equation of the form as in the definition.

1. The system obtained by adding any multiple of any one equation to any second while leaving the other equations unchanged, is equivalent.
2. The system obtained by multiplying any one equation by a nonzero scalar and leaving the other equations unchanged is an equivalent system.
3. The system obtained by interchanging any two equations is an equivalent system.

Proof. Let X be the set of solutions of a system of linear equations as in definition 1.5.1. Then, for all three operations, X is also a solution to the new system of linear equation. That is, for such new set of solutions, $X', X \subseteq X'$.

We then observe the “*reversibility*” of the three operations. In general, for type 1. operation applied by multiplying j^{th} by c and summed by i^{th} row. We can reverse the operation by subtraction of j^{th} by c from i^{th} . Similarly, we can multiply any row by c^{-1} note that $c \neq 0$ and reverse operation 2. Finally, for operation type 3. after interchanging the equation back to its initial we obtain the same set of solution. Thus, $X' \subseteq X$. Therefore, $X = X'$. That is, the set of solution is unique under these three operations, i.e., the system remains equivalent by definition. $\square$

Definition 1.6.3 (Echelon Form System). A system of linear equation in definition 1.5.1 is in *echelon form* if

1. In each equation, all the coefficients are 0 or the leading term is a 1. In the i^{th} equation we will call the subscript of the variable in the leading term $j(i)$.
2. For each i , the coefficient of $x_{j(i)}$ is zero in every equation other than the i^{th} .
3. For each i such that equations $i \wedge i + 1$ have some nonzero coefficient, $j(i + 1) > j(i)$.

Theorem 1.6.3.1. Every system of linear equations is equivalent to a system in echelon form. Furthermore, this echelon form system may be found by applying a sequence of elementary operations to the original system.

Proof. We prove by induction. Let m be the number of equations in a system. We define $P(m)$ as “the theorem holds for system of m equation.”

We want to show $\forall m \geq 1, P(m) \implies P(m + 1)$. First, we show the base case, i.e., $m = 1$. In this case applying operation 2. would suffice for the system to be in echelon form, assuming that coefficients $c \neq 0$. Suppose $c = 0$, the system is by definition already in echelon form. Thus the base case hold.

Now, let $m \geq 1$ be arbitrary. Assume $P(m)$ is true. This means that for a system of m equations, it is equivalent to a system in echelon form obtainable by sequence of elementary operations. Then, for a system of $m + 1$ equations we thus have:

$$\begin{array}{ccccccc} a_{11}x_1 & + & \dots & + & a_{1n}x_n & = & b_1 \\ a_{21}x_1 & + & \dots & + & a_{2n}x_n & = & b_2 \\ & & & & \vdots & & \\ a_{m1}x_1 & + & \dots & + & a_{mn}x_n & = & b_m \\ a_{m+1,1}x_1 & + & \dots & + & a_{m+1,n}x_n & = & b_{m+1} \end{array}$$

Suppose $m + 1$ is not a system $0 = c$ otherwise it is by definition already an echelon form system. We define the nonzero coefficient farthest to the left as a_{ij} . Note that the corresponding variables thus have an index number $\geq j$. As j is the farthest to the left. We then apply operation 3. to swap the first equation and the last equation (the $(m + 1)^{th}$). Then we multiply the new first equation by $\frac{1}{a_{ij}}$ to make the leading coefficient 1. Lastly, by operation 1., we can eliminate any other occurrence of the variable x_j , thus obtaining:

$$\begin{array}{ccccccc} x_j & + & a'_{1,j+1}x_{j+1} & + & \dots & + & a'_{1n}x_n & = & b'_1 \\ & & a'_{2,j+1}x_{j+1} & + & \dots & + & a'_{2n}x_n & = & b'_2 \\ & & & & & & \vdots & & \\ & & a'_{m,j+1}x_{j+1} & + & \dots & + & a'_{mn}x_n & = & b'_m \\ & & a'_{m+1,j+1}x_{j+1} & + & \dots & + & a'_{m+1,n}x_n & = & b'_{m+1} \end{array}$$

where the “'” denotes the coefficient that may be different from the original after operations.

Note that by induction hypothesis, the second to the $m + 1$ equations are equivalent to a system of echelon form. Note that we can eliminate any leading coefficient appears in equation 2 to $m + 1$ in equation 1 by operation 2.. We thus have an echelon form system. By Proposition 1.5.2.1, it is equivalent to our original system of equations. Thus we have demonstrated required by simple induction. $\square$

Definition 1.6.4 (Basic/Free Variables).

1. In an echelon form system the variables appearing in the leading terms of equations are called the *basic variables* of the system
2. All the other variables are called *free variables* (including those that are implicit, i.e., with a coefficient of 0).

Example. Consider the echelon form system:

$$\begin{array}{ccccccccc} x_1 & & & + & x_3 & & & = & 0 \\ & x_2 & & - & x_3 & & + & x_5 & = & 0 \\ & & & & & x_4 & + & x_5 & = & 0 \end{array}$$

Then, the basic variables are x_1, x_2 , and x_4 . The free variables are x_3 and x_5 . Furthermore, the set of solutions when we substitute $x_3 = t_0 \wedge x_5 = t_1$ is:

$$W = \{t_0(-1, 1, 1, 0, 0) + t_1(0, -1, 0, -1, 1) | t_0, t_1 \in \mathbb{R}\}$$

We call this description *parametrization* of the set of solutions. That is, we have defined a mapping $F : \mathbb{R}^2 \rightarrow \mathbb{R}^5$ whose image is the set of solutions. Furthermore, F is given by $F(t_1, t_2) = t_0(-1, 1, 1, 0, 0) + t_1(0, -1, 0, -1, 1) = (-t_1, t_0 - t_1, t_0, -t_1, t_1)$. Geometrically W is a plane containing the origin in the space $V = \mathbb{R}^5$.

Corollary 1.6.4.0.1 (of theorem 1.5.3.1). Every homogeneous system of m linear equations in n unknowns has a nontrivial solution, if $m < n$.

Proof. Let k be the number of basic variables in such linear system. Then, by definition $k \leq m$ since the number of basic variables may be no greater than the number of equations. It then follows that $k \leq m < n \implies k < n$. That is, there are more variables than there are basic variables. Variables can either be basic or free. Since $0 < n - k = \text{number of free variable}$ there is at least a free variable in such linear system. We substitute such free variable with a non-zero real number, the resulting solution is non-trivial. $\square$

Remarks 1.6.4.0.1 (Three motivating problems and applicable solutions).

1. Given a subspace W of $\mathbb{R}^n$ defined as the set of solutions of a system of homogeneous linear equations with n variables m equations, to find a set S s.t. $W = \text{Span}(S)$, we can proceed as follows: First, reduce the system to echelon form. Then the set of solutions may be found exactly as in our previous example. A set S with as many vectors as there are free variables in the echelon form system may be obtained by the sum of, for each free variable in turns, setting that free variable as 1 and all else as 0.
2. Given a finite set of S of vectors in $\mathbb{R}^m$ and $x \in \mathbb{R}^m$, we can determine if $x \in \text{Span}(S)$ by setting up the appropriate system of linear equations and its reductive echelon form. As a rule of thumb, if the system is inconsistent, then $x \notin \text{Span}(S)$; if the system is consistent, then $x \in \text{Span}(S)$ and the scalars in the linear combinations of the vectors in S of the solution vectors yield the vector x .
3. To determine dependency of a set of vectors, we reduce the appropriate homogeneous system of linear equations to echelon form. If the number of free variables, $k \geq 1$, then the system will yield a nontrivial solutions dependent on such free variables. Thus the original system is linearly dependent, as $\exists i$ s.t. $a_i \neq 0$ for an equation $a_1x_1 + \cdots + a_nx_n = 0$. The set of vectors is independent otherwise.

Definition 1.6.5 (Affine Sets). A set of the following form is called an *affine set*:

$$v = \{x_p + \hat{v} : \hat{v} \in \text{Span } S \text{ where } S \text{ is a set of vectors}\}$$

1.7 Bases and Dimension

Definition 1.7.1 (Spanning Set and Basis). A subset $S \subseteq V$ is a *spanning set* of V if $V = \text{Span}(S)$. A set $S \subseteq V$ is a *basis* of V if (i.) S is linearly independent and (ii) S is a spanning set of V .

Theorem 1.7.1.1 (Unique Representation). S is a basis of V if and only if every vector in V has a unique representation.

Theorem 1.7.1.2 (L.I. Extension Lemma). Let $S \subseteq V$ be linearly independent and $\vec{x} \in V$ s.t. $\vec{x} \notin S$. The union $S \cup \vec{x}$ is linearly independent iff $\vec{x} \notin \text{Span}(S)$.

Theorem 1.7.1.3 (Indep. Set to a Basis). Suppose that V has a finite spanning set $T = \{\vec{y}_1, \dots, \vec{y}_n\}$ and $S = \{\vec{x}_1, \dots, \vec{x}_k\}$ is a linearly independent subset of V . There is always a finite basis S' of V such that $S \subseteq S'$.

Theorem 1.7.1.4 (The Dimension Bound). If V has a spanning set $T = \{\vec{y}_1, \dots, \vec{y}_k\}$ and $S = \{\vec{x}_1, \dots, \vec{x}_n\}$ is a linearly independent set, then $n \leq k$.

Theorem 1.7.1.5 (Identical size of Bases). If $S = \{\vec{x}_1, \dots, \vec{x}_k\}$ and $S' = \{\vec{y}_1, \dots, \vec{y}_n\}$ are two finite bases of V , then $n = k$.

Definition 1.7.2 (Dimension). A vector space is *finite dimensional* if it admits a finite basis. The dimension $\dim(V)$ of a finite dimensional vector space is the number of elements in *any* basis of V .

Theorem 1.7.2.1 (Size of L.I. subset and a basis). Let V be a finite-dimensional vector space, where $\dim(V) = n$. Let S be a linearly independent subset of V . Then, S is a basis for V , if $|S| = n$.

2 Linear Transformations

2.1 Linear Transformations

Definition 2.1.1. Let $(V, \boxplus, \boxminus)$ and $(W, \oplus, \odot)$ be two vector spaces. A *linear transformation* (map/ function) is a function $T : V \rightarrow W$ s.t. :

$$T(\vec{u} \boxplus \vec{v}) = T(\vec{u}) \oplus T(\vec{v}) \quad T(a \boxminus \vec{v}) = a \odot T(\vec{v})$$

Theorem 2.1.1.1 (Linear Transformation Preserves Additive Identities). If $T : V \rightarrow W$ is a L.T., then $T(\vec{0}_V) = \vec{0}_W$.

Theorem 2.1.1.2 (Linear Transformations Preserve Addition and Scaling). $T : V \rightarrow W$ is a linear transformation if and only if $T((a \boxminus \vec{u}) \boxplus (b \boxminus \vec{v})) = (a \odot T(\vec{u})) \oplus (b \odot T(\vec{v}))$ for all vectors $\vec{u}, \vec{v} \in V$ and scalars $a, b \in \mathbb{F}$.

Definition 2.1.2. Suppose that $\vec{u} = (u_1, \dots, u_n)$ and $\vec{v} = (v_1, \dots, v_n)$ are vectors in $\mathbb{R}^n$. The *standard inner product* of $\vec{u}$ and $\vec{v}$ is defined to be:

$$\langle \vec{u}, \vec{v} \rangle = \sum_{i=1}^n u_i v_i = u_1 v_1 + \dots + u_n v_n.$$

We define the *length* or *norm* of $\vec{v}$ to be

$$\|\vec{v}\| = \sqrt{\langle \vec{v}, \vec{v} \rangle} = \sqrt{v_1^2 + \dots + v_n^2}$$

with properties:

- i. $\langle \vec{u}, \vec{v} \rangle = \langle \vec{v}, \vec{u} \rangle$
- ii. $\langle \vec{u}, \vec{v}_1 + \vec{v}_2 \rangle = \langle \vec{u}, \vec{v}_1 \rangle + \langle \vec{u}, \vec{v}_2 \rangle$
- iii. $\langle \vec{u}, a\vec{v} \rangle = a\langle \vec{u}, \vec{v} \rangle$
- iv. $\langle \vec{u}, \vec{u} \rangle \geq 0$, with equality if $\vec{u} = \vec{0}$.

Theorem 2.1.2.1 (Inner products define angles). If $\vec{u}$ and $\vec{v}$ are non-zero vectors in $\mathbb{R}^n$ then the angle θ between $\vec{u}$ and $\vec{v}$ satisfies:

$$\cos(\theta) = \frac{\langle \vec{u}, \vec{v} \rangle}{\|\vec{u}\| \|\vec{v}\|}.$$

Theorem 2.1.2.2 (Orthogonality). Two non-zero vectors $\vec{u}$ and $\vec{v} \in \mathbb{R}^n$ form a right angle if and only if $\langle \vec{u}, \vec{v} \rangle = 0$. We say that two such vectors are *orthogonal*.

Theorem 2.1.2.3 (Linear Maps determined by actions on a basis). Suppose that V is a finite dimensional vector space with basis $\alpha = \{\vec{v}_1, \dots, \vec{v}_n\}$. If $S : V \rightarrow W$ and $T : V \rightarrow W$ are linear maps such that $S(\vec{v}_i) = T(\vec{v}_i)$ for each $\vec{v}_i$, then $S(\vec{v}) = T(\vec{v}), \forall \vec{v} \in V$.

Theorem 2.1.2.4 ($T : V \rightarrow W$ Characterized by $\dim(V) \times \dim(W)$ numbers). Let V and W be finite dimensional vector spaces with $\dim(V) = n$ and $\dim(W) = k$. Any linear transformation $T : V \rightarrow W$ is uniquely determined by nk scalars.

Definition 2.1.3 (The Matrix of a Linear Transformation). Let V and W be finite dimensional vector space with $\dim(V) = n$ and $\dim(W) = k$. Fix bases $\alpha = \{\vec{v}_1, \dots, \vec{v}_n\}$ and $\beta = \{\vec{w}_1, \dots, \vec{w}_k\}$ with $V = \text{Span}(\alpha)$ and $W = \text{Span}(\beta)$. Suppose $T : V \rightarrow W$ is a linear transformation with:

$$T(\vec{v}_i) = a_{1i}\vec{w}_1 + a_{2i}\vec{w}_2 + \dots + a_{ki}\vec{w}_k = \sum_{j=1}^k a_{ji}w_j$$

The *matrix* of T with respect to α and β is

$$[T]_{\alpha}^{\beta} = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{k1} & a_{k2} & \dots & a_{kn} \end{bmatrix}$$

where the coefficient in $T(\vec{v}_i)$ occur as the i^{th} column of $[T]_{\alpha}^{\beta}$.

Definition 2.1.4 (Coordinate Vectors). Let $\beta = \{\vec{v}_1, \dots, \vec{v}_n\}$ be a basis of V . Any vector $\vec{v} \in V$ can be written uniquely as

$$\vec{v} = b_1\vec{v}_1 + \dots + b_n\vec{v}_n = \sum_{i=1}^n b_i\vec{v}_i.$$

We define the *coordinate vector* of v with respect to β to be:

$$[\vec{v}]_{\beta} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix}.$$

Definition 2.1.5 (Matrix-Vector Mult.). Given a matrix $[T]_{\alpha}^{\beta}$ and volume vector $[\vec{v}]_{\alpha}$ we define the *product* of $[T]_{\alpha}^{\beta}$ with $[\vec{v}]_{\alpha}$ as follows:

$$[T]_{\alpha}^{\beta}[\vec{v}]_{\alpha} = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{k1} & a_{k2} & \dots & a_{kn} \end{bmatrix} \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix} = \begin{bmatrix} a_{11}b_1 & a_{12}b_2 & \dots & a_{1n}b_n \\ a_{21}b_1 & a_{22}b_2 & \dots & a_{2n}b_n \\ \vdots & \vdots & & \vdots \\ a_{k1}b_1 & a_{k2}b_2 & \dots & a_{kn}b_n \end{bmatrix} = \begin{bmatrix} \sum_{i=1}^n a_{1i}b_i \\ \sum_{i=1}^n a_{2i}b_i \\ \vdots \\ \sum_{i=1}^n a_{ki}b_i \end{bmatrix}$$

where the size of the product is given by $(k \times n) \cdot (n \times m) = km$.

Theorem 2.1.5.1 (Matrix-Vector Mult. in Coordinates). Fix bases $\alpha = \{\vec{v}_1, \dots, \vec{v}_n\}$ of V and $\beta = \{\vec{w}_1, \dots, \vec{w}_k\}$ of W . If $T : V \rightarrow W$ is a linear transformation, then

$$[T(\vec{v})]_{\beta} = [T]_{\alpha}^{\beta}[\vec{v}]_{\alpha}$$

Definition 2.1.6 (The Transformation Defined by a Matrix). Fix bases $\alpha = \{\vec{v}_1, \dots, \vec{v}_n\}$ of V and $\beta = \{\vec{w}_1, \dots, \vec{w}_k\}$ of W . If $M \in \mathbf{M}_{k \times n}(\mathbb{R})$, then the transformation defined by M is $T_M : V \rightarrow W$ s.t. :

$$T_M(\vec{v}) = \vec{w} \iff M[\vec{v}]_{\alpha} = [\vec{w}]_{\beta}$$

Theorem 2.1.6.1 (The Fundamental Theorem of Matrices).

- i. If $M \in \mathbf{M}_{k \times n}(\mathbb{R})$ then T_M is a linear transformation.
- ii. If $T : V \rightarrow W$ has matrix $M = [T]_{\alpha}^{\beta}$, then $T = T_M$.
- iii. If $M \in \mathbf{M}_{k \times n}(\mathbb{R})$ then $M = [T_M]_{\alpha}^{\beta}$.

Example. Rotation Consider the linear map $R_\theta : \mathbb{R}^2 \rightarrow \mathbb{R}^2$:

$R_\theta(\vec{v}) =$ the vector rotated counter-clockwise about the origin by θ radians.

Let

$$\vec{v} = (x, y) = (\|\vec{v}\| \cos(\phi), \|\vec{v}\| \sin(\phi)). \quad (1)$$

$$\vec{w} = R_\theta(\vec{v}) = (\|\vec{v}\| \cos(\phi + \theta), \|\vec{v}\| \sin(\phi + \theta)). \quad (2)$$

Note that by double angel formula,

$$\cos(\phi + \theta) = \cos(\phi) \cos(\theta) - \sin(\phi) \sin(\theta) \quad (3)$$

$$\sin(\phi + \theta) = \sin(\phi) \cos(\theta) + \cos(\phi) \sin(\theta). \quad (4)$$

Thus,

$$\begin{aligned} R_\theta(\vec{v}) &= (\|\vec{v}\| [\cos(\phi) \cos(\theta) - \sin(\phi) \sin(\theta)], \|\vec{v}\| [\sin(\phi) \cos(\theta) + \cos(\phi) \sin(\theta)]) \\ &= (x \cos(\theta) - y \sin(\theta), y \cos(\theta) + x \sin(\theta)). \end{aligned} \quad \text{by (1)}$$

Example. Projection. Let $\mathbf{a} \in \mathbb{R}^2$ be a non-zero vector. Let a line $L = \text{Span}\{\mathbf{a}\} \subset \mathbb{R}^2$. Consider a map $P_a : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ s.t.

$P_a(\mathbf{v}) =$ given $\vec{v} \in \mathbb{R}^2$, construct a perpendicular line to L through $\vec{v}$ which intersects L at $P_a(\mathbf{v})$.

Recall the orthogonality theorem. The perpendicularity implies that $\langle \mathbf{a}, (\vec{v} - \vec{w}) \rangle = 0$. Where $\vec{w} \in L \implies \vec{w} = c\vec{a}$, for some c . Thus,

$$\begin{aligned} \langle \mathbf{a}, (\vec{v} - \vec{w}) \rangle &= 0 \\ \implies \langle \mathbf{a}, (\vec{v} - c\mathbf{a}) \rangle &= 0 \\ \implies \langle \mathbf{a}, \vec{v} \rangle - c\langle \mathbf{a}, \mathbf{a} \rangle &= 0 \\ \implies c &= \frac{\langle \mathbf{a}, \vec{v} \rangle}{\langle \mathbf{a}, \mathbf{a} \rangle} \end{aligned}$$

That is,

$$P_a(\mathbf{v}) = c\mathbf{a} = \frac{\langle \mathbf{a}, \vec{v} \rangle}{\langle \mathbf{a}, \mathbf{a} \rangle} \mathbf{a}.$$

2.2 Kernel and Image

Definition 2.2.1 (Kernel). The *kernel* (or *nullspace*) of a linear map $T : V \rightarrow W$ is:

$$\ker(T) = \{\vec{v} \in V : T(\vec{v}) = \vec{0}_W\}.$$

Theorem 2.2.1.1 (The Kernel is a subspace). For any linear map $T : V \rightarrow W$, the kernel $\ker(T)$ is a subspace of V .

Definition 2.2.2 (Image). The *image* of a linear map $T : V \rightarrow W$ is

$$\text{image}(T) = \{\vec{w} \in W : \vec{w} = T(\vec{v}) \text{ for some } \vec{v} \in V\}.$$

Theorem 2.2.2.1 (The Image is a Subspace). For any linear map $T : V \rightarrow W$, the image $\text{image}(T)$ is a subspace of W .

Theorem 2.2.2.2 (Kernels in Coordinates). Kernels are solutions of homogeneous linear systems.

$$T(\vec{x}) = \vec{0} \iff [T]_{\alpha}^{\beta}[\vec{x}]_{\alpha} = [\vec{0}]_{\beta}$$

Theorem 2.2.2.3 (Image and Spans). If $\{\vec{v}_1, \dots, \vec{v}_n\}$ is a spanning set for V and $T : V \rightarrow W$ is a linear transformation, then $\{T(\vec{v}_1), T(\vec{v}_2), \dots, T(\vec{v}_n)\}$ is a spanning set for $\text{image}(T)$.

Theorem 2.2.2.4 (Image and Columns). Fix bases $\alpha = \{\vec{v}_1, \dots, \vec{v}_n\}$ and $\beta = \{\vec{w}_1, \dots, \vec{w}_k\}$ of V and W . The $\text{image}(T)$ is spanned by the columns of $[T]_{\alpha}^{\beta}$ which contain basic variables of the homogeneous system $[T]_{\alpha}^{\beta} = \vec{0}$.

Theorem 2.2.2.5 (Dimension Theorem). If V and W are finite dimensional vector spaces and $T : V \rightarrow W$ is linear, then, $\dim(\ker(T)) + \dim(\text{image}(T)) = \dim(V)$.

Theorem 2.2.2.6 (Bases for image and kernel). If V and W are finite dimensional vector spaces and $\dim(V) = n$ then there is a basis:

$$V = \text{Span}\{\vec{u}_1, \dots, \vec{u}_k, \vec{v}_{k+1}, \dots, \vec{v}_n\}$$

such that $\ker(T) = \text{Span}\{\vec{u}_1, \dots, \vec{u}_k\}$ and $\text{image}(T) = \text{Span}\{T(\vec{v}_{k_1}), \dots, T(\vec{v}_n)\}$.

Theorem 2.2.2.7 (Injective and Surjective). A linear mapping $T : V \rightarrow W$ is injective (one-one) if

$$T(\vec{x}) = T(\vec{y}) \implies \vec{x} = \vec{y}.$$

Else, surjective (onto) if

$$\forall \vec{w} \in W, \exists \vec{v} \in V \text{ s.t. } T(\vec{v}) = \vec{w}.$$

Theorem 2.2.2.8 (kernels and injectivity). The followings are equivalent:

1. T is injective
2. $\ker(T) = \{\vec{0}_V\}$
3. $\dim(\ker(T)) = 0$.

Theorem 2.2.2.9 (image and surjectivity). For a linear transformation $T : V \rightarrow W$ where W is finite-dimensional vector space. The followings are equivalent:

1. T is surjective.
2. $\text{image}(T) = W$.
3. $\dim(\text{image}(T)) = \dim(W)$.

Theorem 2.2.2.10 (Dimension and injectivity). Given a linear transformation $T : V \rightarrow W$. If $\dim(V) > \dim(W)$, then T is *not* injective. Equivalently, if T is injective, then $\dim(V) \leq \dim(W)$.

Theorem 2.2.2.11 (Dimension and surjectivity). Given a linear transformation $T : V \rightarrow W$. If $\dim(V) < \dim(W)$, then T is *not* surjective. Equivalently, if T is surjective, then $\dim(V) \geq \dim(W)$.

Theorem 2.2.2.12 (Finite bijection). For $T : V \rightarrow W$ and $\dim(V) = \dim(W) = n < \infty$, T is injective iff T is surjective.

2.3 Composition and Inverse

Definition 2.3.1 (Composition). Two functions, $T : U \rightarrow V$ and $S : V \rightarrow W$ has their composition defined:

$$ST : U \rightarrow W \equiv ST(\vec{x}) = S(T(\vec{x})).$$

The notation $ST \equiv S \circ T$, i.e., S of T . For the composition to be defined, inputs and outputs must satisfy:

$$U \xrightarrow{T} V \xrightarrow{S} W \text{ s.t. } \text{image}(T) \subseteq \text{dom}(S).$$

Theorem 2.3.1.1 (Linear composition of linear maps). If $T : U \rightarrow V$ and $S : V \rightarrow W$ are linear, then

$$ST : U \rightarrow W \text{ is linear.}$$

Proof. Suppose T and S are linear and their composition $ST : U \rightarrow W$ is defined. Let $\vec{x}, \vec{y} \in U$ and $a, b \in \mathbb{F}$. Then,

$$\begin{aligned} ST(a\vec{x} + b\vec{y}) &= S(aT(\vec{x}) + bT(\vec{y})) && T \text{ is linear} \\ &= aS(T(\vec{x})) + bS(T(\vec{y})) && S \text{ is linear} \\ &= aST(\vec{x}) + bST(\vec{y}) && \text{definition of } ST. \end{aligned}$$

Thus, $ST : U \rightarrow W$ is linear. □

Definition 2.3.2 (Adding and Scaling). Suppose that $T : V \rightarrow W$ and $S : V \rightarrow W$ are linear transformations. Their sum is defined:

$$(S + T)(\vec{v}) = S(\vec{v}) + T(\vec{v})$$

Theorem 2.3.2.1 (Nice Properties of composition).

1. Composition is associative. Consider three linear maps:

$$R : U \rightarrow V \quad S : V \rightarrow W \quad T : W \rightarrow X$$

$$\text{then } (TS)R = T(SR).$$

2. Composition is distributive. Consider

$$R : U \rightarrow V \quad S : V \rightarrow W \quad T : V \rightarrow W$$

$$\text{then } (T + S)R = TR + SR.$$

Theorem 2.3.2.2 (Kernal, Images, and Composition). Suppose $S : U \rightarrow V$ and $T : V \rightarrow W$ are linear maps:

1. $\ker(S) \subseteq \ker(TS)$
2. $\text{image}(TS) \subseteq \text{image}(T)$.

Definition 2.3.3 (Matrix Product (important)). Fix basis $\alpha = \{\vec{u}_1, \dots, \vec{u}_m\}, \beta = \{\vec{v}_1, \dots, \vec{v}_k\}, \gamma = \{\vec{w}_1, \dots, \vec{w}_l\}$ of U, V, W . If $S : U \rightarrow V$ and $T : V \rightarrow W$ then we define the *product* of $[T]_\beta^\gamma$ and $[S]_\alpha^\beta$ as follows:

$$[T]_\beta^\gamma [S]_\alpha^\beta = [TS]_\alpha^\gamma.$$

Theorem 2.3.3.1 (Matrix Mult Formula). If $T = [t_{ij}] \in \mathbf{M}_{l \times k}(F)$ and $S = [s_{ij}] \in \mathbf{M}_{k \times n}(F)$ then this gives $TS = [p_{ij}]$ where

$$p_{ij} = \sum_{N=1}^k t_{iN} s_{Nj}$$

Definition 2.3.4 (An Inverse of a Function). Suppose that $S : V \rightarrow W$ is a function, an inverse of S is $T : W \rightarrow V$ s.t.

$$TS(\vec{v}) = \vec{v}, \forall \vec{v} \in V \text{ and } ST(\vec{w}) = \vec{w}, \forall \vec{w} \in W.$$

Theorem 2.3.4.1 (Injectivity, Surjectivity, Inverse). A function $S : V \rightarrow W$ has an inverse $T : W \rightarrow V$ if and only if S is injective and surjective.

Theorem 2.3.4.2 (Inverses are Unique). If a function $S : V \rightarrow W$ has an inverse $T : W \rightarrow V$ then T is unique.

Theorem 2.3.4.3 (The inverse is linear). If a linear transformation $S : V \rightarrow W$ has an inverse $T : W \rightarrow V$, then

T is linear.

Theorem 2.3.4.4 (Row Operations are Invertible). Every row operation $M \xrightarrow{X} M'$ can be represented by an invertible matrix E_X such that

$$M' = E_X M.$$

Definition 2.3.5 (Isomorphism). An invertible linear transformation $T : V \rightarrow W$ is an isomorphism

Theorem 2.3.5.1 (Iso and Injectivity/Surjectivity). A linear transformation $T : V \rightarrow W$ is an isomorphism iff it is injective and surjective

Remark. When $\dim(V) = \dim(W)$ it suffices to check either injec or surjec.

Definition 2.3.6 (Automorphism). An invertible linear transformation $T : V \rightarrow V$ is an automorphism.

Axiom 2.3.7 (Axioms of Group). Group has three axioms:

A_G1 . There is an associative multiplication operation

$$G \times G \rightarrow G.$$

A_G2 . There is an identity element $e \in G$ s.t.

$$eg = ge = g.$$

A_G3 . For any $g \in G$, there is an inverse g^{-1} such that

$$gg^{-1} = g^{-1}g = e.$$

Theorem 2.3.7.1 (The Gauss-Jordan Algorithm for Finding Inverse). Input: A matrix $M \in M_{n \times n}(\mathbb{R})$. Output: If M has an inverse, output M^{-1} .

1. From the augmented matrix $[M|I]$ with $n \times n$ identity.
2. Reduce M in $[M|I]$ to EF and obtain $[A|M']$.
3. If $A = I$ then $M' = M^{-1}$. Otherwise, M is not invertible.

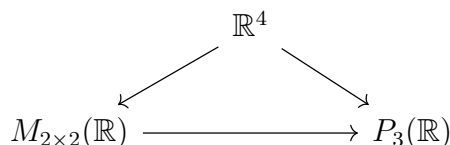
Example. Prove $M_{2 \times 2}(\mathbb{R})$ and $\mathbb{R}^4$ are isomorphic.

Proof. Consider $T\left(\begin{bmatrix} a & b \\ c & d \end{bmatrix}\right) = (a, b, c, d)$. Note that $T : M_{2 \times 2}(\mathbb{R}) \rightarrow \mathbb{R}^4$ is linear, surjective, and injective. Therefore it suffices as an isomorphism. $\square$

Example. Prove $P_3(\mathbb{R})$ and $\mathbb{R}^4$ are isomorphic.

Proof. Consider $S(ax^3 + bx^2 + cx + d) = (a, b, c, d)$. Again as S is linear, injective, and surjective it is an isomorphism. $\square$

Remark.



Theorem 2.3.7.2 (Classification of Finite Dimensional Vector Spaces). If V is an n -dimensional vector space defined over the field $\mathbb{F}$, then V is isomorphic to $\mathbb{F}^n$.

Corollary 2.3.7.2.1. Two finite-dimensional real vector spaces V and W are isomorphic if and only if $\dim(V) = \dim(W)$.

Proof. Suppose V has a basis $\{\vec{v}_1, \dots, \vec{v}_n\}$. Let $T : V \rightarrow \mathbb{F}^n$ such that $T(x_1\vec{v}_1 + \dots + x_n\vec{v}_n) = (x_1, \dots, x_n)$. Note that this mapping is linear, injective, and surjective. Therefore it is an isomorphism. $\square$

Proof of Corollary. Suppose V and W are isomorphic. There is a $T : V \rightarrow W$ such that T is linear, surjective, and injective. As T is surjective, $\dim(V) \geq \dim(W)$. As T is injective, $\dim(V) \leq \dim(W)$. Thus $\dim(W) = \dim(V)$. For the other direction, suppose $\dim(V) = \dim(W) = n$. By classification, there is there is an isomorphism $T : V \rightarrow \mathbb{R}^n$. Similarly there is an isomorphism $S : W \rightarrow \mathbb{R}^n$. The composition $S^{-1}T : V \rightarrow W$ is an isomorphism. Therefore V and W are isomorphic vector spaces. $\square$

3 Change of Basis / Determinants

Definition 3.0.1 (Alternating and Multilinear). A function $T : V^n \rightarrow W$ is *multilinear* if it is linear in each of its n inputs:

$$T(\vec{v}_1, \dots, a\vec{v}_i + b\vec{v}_i', \dots, \vec{v}_n) = aT(\vec{v}_1, \dots, \vec{v}_i, \dots, \vec{v}_n) + bT(\vec{v}_1, \dots, \vec{v}_i', \dots, \vec{v}_n).$$

A function $T : V^n \rightarrow \mathbb{R}$ is *alternating* if switching the order of two entries changes the sign:

$$T(\vec{v}_1, \dots, a\vec{v}_i, \dots, \vec{v}_j, \dots, \vec{v}_i, \dots, \vec{v}_n) = -T(\vec{v}_1, \dots, a\vec{v}_j, \dots, \vec{v}_i, \dots, \vec{v}_n)$$

Theorem 3.0.1.1. If function $T : V^n \rightarrow \mathbb{R}$ is alternating and has two identical inputs, then it outputs zero:

$$T(\vec{v}_1, \dots, a\vec{v}_i, \dots, \vec{v}_i, \dots, \vec{v}_n) = 0.$$

Theorem 3.0.1.2 (Detecting Dependence). If $T : V^n \rightarrow \mathbf{F}$ is *alternating* and *multilinear* and the set of input vectors is linearly dependent, then the output will be zero.

Theorem 3.0.1.3 (The characterization of the det function). Consider $M_{n \times n}(\mathbb{R})$ as $(\mathbb{R}^n)^n$ where the rows of the matrix are thought of as vectors in $\mathbb{R}^n$. If $f : M_{n \times n}(\mathbb{R}) \rightarrow \mathbb{R}$ is alternating, multilinear, and satisfies $f(I) = 1$, then $f = \det$.

Theorem 3.0.1.4 (Important- Determinants and invertibility). A matrix $M \in M_{n \times n}(\mathbb{R})$ is invertible if and only if $\det(M) \neq 0$,

Theorem 3.0.1.5. Let $M \in M_{n \times n}(\mathbb{R})$. Suppose that M' is obtained from M by row operation.

Scale a row. If $M(\lambda R_i \rightarrow R_i) \rightarrow M'$ then $\det(M') = \lambda \det(M)$.

Exchange R_i and R_j . If $M \xrightarrow{R_i \leftrightarrow R_j} M'$ then $\det(M') = -\det(M)$.

Add a multiple of a row. If $M(R_j + \lambda R_i \rightarrow R_j) \rightarrow M'$ then $\det(M') = \det(M)$.

Note: These properties of det are sometimes called “row operation invariance”. This is not quite right, because the determinant does change when applying row operations. However, the determinant changes in simple and predictable ways.

Definition 3.0.2 (Submatrices - Minors). The $i - j$ -minor of $M \in M_{n \times n}(\mathbb{R})$ is the matrix M_{ij} obtained by deleting row i and column j from M .

Theorem 3.0.2.1 (Recursive Formula for det). If $M = [m_{ij}] \in M_{n \times n}(\mathbb{R})$ and $1 \leq i \leq n$ then we define:

$$\det(M) = \sum_{j=1}^n (-1)^{i+j} m_{ij} \det(M_{ij}) = \sum_{i=1}^n (-1)^{i+j} m_{ij} \det(M_{ij})$$

Remark. see lecture note week 9 page 11 example.

Theorem 3.0.2.2 (Changing Coordinates). Sps V is a finite-dimensional vector space with bases α and β . Let $I : V \rightarrow V$ be the identity transformation. The coordinate vectors of $[\vec{v}]_\alpha$ and $[\vec{v}]_\beta$ are related such that

$$[I]_\alpha^\beta [\vec{v}]_\alpha = [\vec{v}]_\beta.$$

We call $[I]_\alpha^\beta$ the change of basis matrix from $\alpha \rightarrow \beta$.

Theorem 3.0.2.3 (Changing Coordinates and Linear Maps). Sps $T : V \rightarrow W$ is a linear map given as a matrix $[T]_\alpha^\beta$. If V has two bases α and α' and W has two bases β and β' then

$$[T]_{\alpha'}^{\beta'} = [I_W]_\beta^{\beta'} [T]_\alpha^\beta [I_V]_{\alpha'}^\alpha$$

Theorem 3.0.2.4 (A useful special case- changing coordinates of $T : V \rightarrow V$).

$$[T]_{\alpha'}^{\alpha'} = [I]_\alpha^{\alpha'} [T]_\alpha^\alpha [I]_{\alpha'}^\alpha$$

Definition 3.0.3 (n-fold composition). If we have a linear transformation $T : V \rightarrow V$ we can consider applying it repeatedly. We write $T^n(\vec{v}) = T(T(\dots T(\vec{v})))$ where T is applied n times. This defined the *n-fold composition* of T .

4 Eigenvalues and Eigenvectors

Definition 4.0.1 (Eigenvalues and Eigenvectors). Suppose we have a linear map $T : V \rightarrow V$ s.t. $T(x) = \lambda x$ for some non-zero $x \in V$. We say that x is *eigenvector* with *eigenvalue* λ .

Theorem 4.0.1.1 (Eigenvectors Are vectors in kernels).

$$T(\vec{v}) = \lambda \vec{v} \iff \vec{v} \in \ker(T - \lambda I).$$

Proof.

$$\begin{aligned} T(\vec{v}) = \lambda \vec{v} &\iff T(\vec{v}) = \lambda I(\vec{v}) && \text{identity} \\ &\iff T(\vec{v}) - \lambda I(\vec{v}) = \vec{0} && \text{additive inverse} \\ &\iff (T - \lambda I)(\vec{v}) = \vec{0} && \text{as } \mathcal{L}(V, V) \text{ is a vector space} \\ &\iff \vec{v} \in \ker(T - \lambda I) && \text{definition of kernel} \end{aligned}$$

□

Definition 4.0.2 (The Characteristic Polynomial). The *characteristic polynomial* of a matrix A is $\chi_A(\lambda) = \det(A - \lambda I)$.

Theorem 4.0.2.1 (Eigenvalues of the roots of the characteristic equation). A scalar λ is an eigenvalue of T if and only if it is a root to $\chi_{[T]_\alpha}^\alpha$

Definition 4.0.3 (Eigenvalues and Eigenvectors of a matrix). Let $M \in M_{n \times n}(\mathbb{F})$. Let $x \in M_{n \times 1}(\mathbb{F})$ be a non-zero vector. Then x is an eigenvector of eigenvalue $\lambda \in \mathbb{F}$ if

$$Mx = \lambda x.$$

Theorem 4.0.3.1 (Eigenvalues are the roots of the characteristic poly). Let $M \in M_{n \times n}(\mathbb{F})$. Then, the eigenvalues of M are the roots of $\chi_M(x)$, where

$$\chi_M(x) = \det(M - \lambda I).$$

Definition 4.0.4 (λ - Eigenspace E_λ). If $T : V \rightarrow V$ is a linear transformation then its λ -*eigenspace* is

$$E_\lambda = \{\vec{v} : T(\vec{v}) = \lambda \vec{v}\}.$$

Theorem 4.0.4.1 (E_λ is a subspace). For any $\lambda \in \mathbb{F}$, we have that E_λ is a subspace of V .

Theorem 4.0.4.2 (Determinants and Eigenvalues). If A has eigenvalues λ_i s.t. $i \in [1, n] \cap \mathbb{N}$, then

$$\det(A) = \prod_{i=1}^n \lambda_i$$

Theorem 4.0.4.3 (Invertibility and Eigenvalues). A is invertible iff $\lambda = 0$ is not an eigenvalue.

5 Diagonalizability

Theorem 5.0.0.1 (LADR find dependence). Suppose that V is n -dimensional real vector space. If $T : V \rightarrow V$ is a linear transformation, then $\{I, T, \dots, T^{n \cdot n}\}$ is linearly dependent in $\mathcal{L}(V, V)$. Alternatively, there are non-zero constants such that

$$Z = a_{n \cdot n} T^{n \cdot n} + \dots + a_2 T^2 + a_1 T^1 + a_0 I$$

where $I : V \rightarrow V$ is identity and $Z : V \rightarrow V$ is the zero map.

Proof. Note that if $T : V \rightarrow V$ is linear, then $T^k : V \rightarrow V$ is linear for $k \geq 1$. We have

$$\dim(\mathcal{L}(V, V)) = n^2.$$

Consider

$$\{I, T, \dots, T^{n \cdot n}\} \subseteq \mathcal{L}(V, V).$$

Thus set must be dependent as $|\{I, T, \dots, T^{n \cdot n}\}| = n^2 + 1 > n^2$. Which yields non-zero constant

$$Z = a_{n \cdot n} T^{n \cdot n} + \dots + a_2 T^2 + a_1 T^1 + a_0 I$$

as needed. □

Theorem 5.0.0.2 (Fundamental Theorem of Algebra). If $p(x) \in P_n(\mathbb{R})$ is a degree n polynomial with real coefficients of the form:

$$p(x) = x^n + a_{n-1}x + d + a_0$$

then $p(x)$ factors uniquely in the form:

$$p(x) = (x - \lambda_1)^{m_1} (x - \lambda_2)^{m_2} \dots (x - \lambda_k)^{m_k}$$

where $m_1 + \dots + m_k = n \wedge \lambda_i \in \mathbb{C}$.

Theorem 5.0.0.3 (Bound on number of eigenvalues). If $M \in \mathbb{M}_{n \times n}(\mathbb{R})$, then M has at most n real eigenvalues.

Definition 5.0.1 (Diagonalizability). A linear transformation $T : V \rightarrow V$ of a finite-dimensional vector space is *diagonalizable* if there is a basis α of V consisting of eigenvectors of T .

Theorem 5.0.1.1 (Characterization of Diagonalizability). $T : V \rightarrow V$ is diagonalizable if and only if $[T]_\alpha^\alpha$ is *similar* to a diagonal matrix.

Theorem 5.0.1.2 (Eigenvectors of Distinct Eigenvalues are Independent). Suppose that $T(\vec{v}_i) = \lambda_i \vec{v}_i$ for $i = 1, \dots, n$ are distinct eigenvalues and eigenvectors of T . The set of vectors $\{\vec{v}_1, \dots, \vec{v}_n\}$ is linearly independent.

Theorem 5.0.1.3 (The Nice Case of Diagonalization). Suppose that V is an n -dimensional real vector space. If $T : V \rightarrow V$ has n distinct real eigenvalues, then T is diagonalizable.

Definition 5.0.2 (Algebraic and Geometric Multiplicity). The *algebraic multiplicity* $\text{Alg}(\lambda)$ of an eigenvalue λ is the number of times that it occurs as a root of the characteristic polynomial of T . The *geometric multiplicity* $\text{Geo}(\lambda)$ of an eigenvalue λ is the $\dim(E_\lambda)$.

Theorem 5.0.2.1 (Geometry $\leq$ Algebra). Suppose that V is an n -dimensional real vector space. If $T : V \rightarrow V$ has a real eigenvalue λ , then $\text{Geo}(\lambda) \leq \text{Alg}(\lambda)$.

Theorem 5.0.2.2 (Indep. Sets from Eigenspaces). Suppose that V is an n -dimensional vector space with distinct real eigenvalues λ_1 and λ_2 . If $S_1 \subset E_{\lambda_1}$ and $S_2 \subset E_{\lambda_2}$ are sets of independent vectors, then $S_1 \cup S_2$ is independent.

Theorem 5.0.2.3 (The General Case of Diagonalization). Suppose that $T : V \rightarrow V$ is a linear transformation of a finite-dimensional vector space. Further suppose that it has distinct real eigenvalues $\lambda_1, \dots, \lambda_k$. Let m_i be the algebraic multiplicity of the eigenvalue λ_i . T is diagonalizable iff:

1. $\dim(V) = m_1 + \dots + m_k = \text{Alg}(\lambda_1) + \dots + \text{Alg}(\lambda_k)$
2. $\dim(E_{\lambda_i}) = m_i$, i.e., $\text{Geo}(\lambda_i) = \text{Alg}(\lambda_i), \forall i$